

ASHER BENNETT

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EDUCATION

Bachelor of Science in Mechanical Engineering

Expected June 2028

University of Massachusetts Amherst | GPA: 3.6/4.0 | Dean's List

Relevant Coursework: Thermodynamics, Fluid Mechanics*, Heat Transfer*, Modeling & Analysis of Dynamic Systems*, Dynamics, Design of Mechanical Components, Manufacturing Processes

* Upcoming coursework

EXPERIENCE

R&D Systems Hardware Engineering Co-op

July 2026 – Present

Werfen (Instrumentation Laboratory), Bedford, MA

- Ran a three-week thermal qualification test on two instrument lines, working through 15 test configurations across ambient, hot, and cold conditions with fans on and off.
- Verified performance against 15 system-level temperature requirements and found 3 subsystems running outside limits at specific conditions, triggering 3 retests and giving the mechanical design team enough data for a redesign.
- Built and calibrated 120 thermistors against a thermal bath to be used in thermal testing, and developed a program to quickly evaluate and determine if each thermistor's performance lied within the acceptable range
- Qualified 12 pumps against design spec across two pre-production instruments, running 140+ weighed measurements to confirm their accuracy before the manufacturing prototype.
- Produced the flow-rate calibration curves the team uses to tune fluid delivery, sweeping duty cycles across their full range in the instrument's embedded Linux environment.
- Caught 8 fluid-level sensors reading outside limits while checking 24 across supply, cleaning-solution, and waste containers, and successfully presented results to engineers with suggested next steps.
- Kept development units running through hardware bring-up by reprogramming and swapping embedded microcontroller control boards.

Information Technology Lead

April 2024 – Oct 2025

Natick Public Schools, Natick, MA

- Led a team of 5 technicians in diagnosis, repair, and turnaround of 2,000+ student devices (desktops, laptops, tablets) across the district.
- Developed hardware troubleshooting workflows that reduced average device turnaround time by ~1 hour and improved team throughput.

ENGINEERING PROJECTS

UMass Rocketry | Mechanical Payload Lead

Sept 2024 – Sept 2026

- Led mechanical design of the payload across two NASA Student Launch cycles, keeping each part inside the rocket's mass and volume limits and able to survive launch G-forces.
- Built a post-landing soil sampling mechanism: a motorized nose cone that digs a sample, seals it, and runs pH and chemical tests on site.
- Helped the team place top 10 nationally out of 70+ university teams.

ICARUS | Autonomous RC Aircraft

May 2023 – Oct 2023

- Designed, built, and flew a GPS-guided autonomous aircraft from scratch, including the foamboard airframe and a 3D-printed enclosure for the electronics.
- Soldered and flashed a SpeedyBee F405 Wing flight controller and wired the full electronics stack (power regulation, overcurrent protection, GPS) to run ArduPilot/iNav in autonomous flight.

FPGA Digital Design | Self-Directed Study

Aug 2026 – Present

- Teaching myself digital design in VHDL on a Basys 3 FPGA, writing the modules and pin constraints by hand and taking each design through Vivado onto hardware.

TECHNICAL SKILLS

Thermal & Fluids: Thermistor assembly & calibration, thermocouples, thermal bath calibration, temperature mapping, pump flow characterization, gravimetric measurement, calibration curve development

Test & Instrumentation: Agilent 34972A DAQ, two-wire resistance measurement, environmental chamber testing, sensor integration & characterization, test protocol execution, Minitab, Excel

CAD & Prototyping: SOLIDWORKS, 3D printing (FDM, multiple materials), Orca/Creality Slicer, rapid prototyping & iteration

Software: Python, C++, MATLAB, Linux, embedded systems (STM32, ArduPilot, SpeedyBee), VHDL, SAP

Manufacturing: GD&T, additive manufacturing, manual machining, electronics soldering & assembly